

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

CLASS

BIOLOGY

8875/01

Paper 1 Multiple Choice

19 September 2014

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and class on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **17** printed pages.

[Turn over

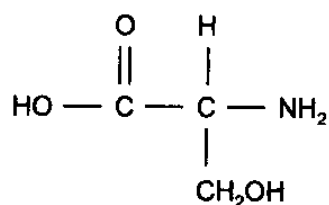
- 1 A certain organelle in a eukaryotic cell was isolated and analysed. It was found that the organelle contains proteins, nucleotides and phospholipids.

Which organelle(s) could it possibly be?

- 1 nucleus
- 2 lysosome
- 3 ribosome
- 4 mitochondrion

- A 1 only
- B 2 and 3
- C 1 and 4
- D 1, 3 and 4

- 2 The diagram below shows a molecule.



Which substance below will contain this molecule?

- A Amylase
- B Amylose
- C Glycogen
- D Starch

- 3 Which statement is true for phospholipids, but not for protein?

- A It has hydrophilic and hydrophobic components.
- B It is synthesized from non-identical sub-units.
- C It can form a barrier to water soluble molecules.
- D It is found in cell membranes.

- 4 Myoglobin is a single-chain globular protein with a haem prosthetic group. The following diagram shows the structure of myoglobin:



Myoglobin can be denatured under certain conditions. Which of the following options correctly match the denaturing conditions with their effects on the protein?

	denaturing condition	level(s) of protein structure disrupted	type(s) of bond(s) primarily affected
A	large change in pH	secondary, tertiary and quaternary	ionic bonds
B	addition of organic solvents	primary, secondary and tertiary	hydrophobic interactions
C	raising the temperature to 70°C	secondary, tertiary	hydrogen bonds, hydrophobic interactions, disulphide bonds
D	raising the temperature to 60°C	secondary, tertiary	ionic bonds, hydrogen bonds, hydrophobic interactions

- 5 Equal quantities of a 5% starch solution were added to equal quantities of different solutions.

After 5 minutes at room temperature, the solutions were tested with iodine in potassium iodide solution and the depth of colour was recorded using a colorimeter (with an arbitrary scale of 1 to 5, 5 being the darkest blue).

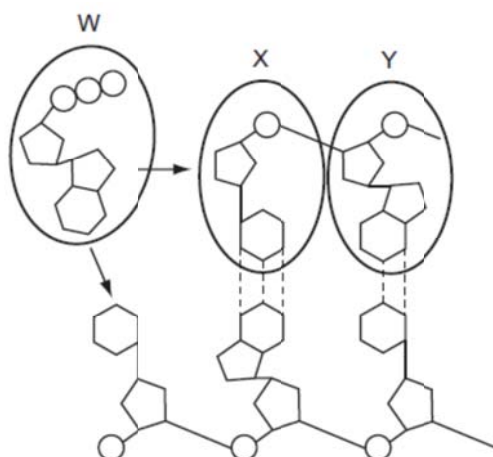
The results are given in the table.

solution	colorimeter reading
1 saliva and dilute HCl	4
2 saliva and dilute NaHCO ₃	1
3 dilute HCl only	4
4 saliva at 100°C	5
5 saliva at 20°C	3

This experiment suggests that saliva contains a biologically active substance that

- A hydrolyses starch in acid conditions
 - B is denatured by hydrochloric acid
 - C is promoted in its action by high temperatures
 - D may be activated by hydrogen carbonate ions
- 6 When cytoplasm from a cell that is undergoing mitosis is injected into another cell that is in interphase, the cell receiving the cytoplasm
- A terminates in interphase
 - B enters mitosis
 - C undergoes apoptosis (cell death)
 - D becomes haploid

- 7 The diagram shows the synthesis of a polynucleotide. Molecule W is a nucleotide triphosphate.



Which statements are correct?

- 1 The base in W could be the purine, adenine
 - 2 The base in Y is the purine, guanine
 - 3 The base in X is the pyrimidine, cytosine
 - 4 The base in X could be the pyrimidine, uracil
- A** 1 and 3
B 2 and 3
C 2 and 4
D all of the above
- 8 The table below shows a list of characteristics displayed by mutant strains of *E. coli* during DNA replication and the possible reasons.

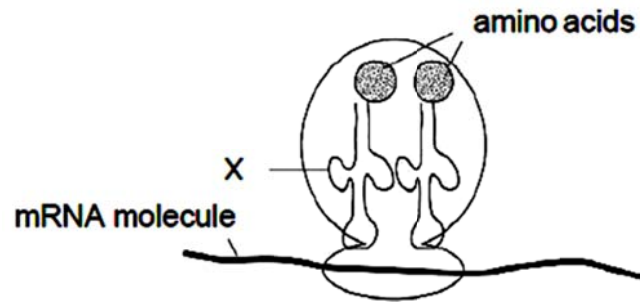
No.	Characteristics of mutant <i>E. coli</i>	Enzymes or functions affected by mutation
1	Okazaki fragments accumulate and DNA synthesis is not complete	DNA ligase activity is missing
2	Parental DNA molecule does not unzip	DNA helicase is hyperactive
3	Synthesis of complementary strands are slow	DNA polymerase keeps dissociating from the DNA and has to re-associate
4	No initiation of replication occurs	A-T rich region at origin of replication deleted

Which of the reasons correctly explain the characteristics displayed by the mutant *E. coli* strains?

- A** 1 and 3
B 2 and 3
C 1, 3 and 4
D all of the above

- 9 If a protein is made up of 30 amino acids and DNA only contained three different types of nitrogenous bases, what would be the minimum number of bases per codon that could code for this protein?
- A 3
 - B 4
 - C 6
 - D 8
- 10 Which statements about the genetic code are correct?
- 1 The genetic code is degenerate.
 - 2 There is only one codon for the amino acid methionine.
 - 3 Codons act as 'stop' and 'start' signals during transcription and translation.
 - 4 Prokaryotes generally use the same genetic code as eukaryotes.
- A 1 and 2
 - B 1, 2 and 3
 - C 1, 2 and 4
 - D all of the above
- 11 5' – CAU – 3' is a codon in mRNA that specifies the amino acid histidine (His) in position 58 in the α chain of haemoglobin. What is the corresponding anti-codon in tRNA?
- A 5' – CAU – 3'
 - B 3' – AUG – 5'
 - C 3' – GUA – 5'
 - D 5' – GUA – 3'

12 The diagram below shows part of a molecule of mRNA bound to a ribosome.



Which of the following is **false** about molecule X?

- 1 It is formed by RNA polymerase in the nucleus.
- 2 It is able to form hydrogen bonds with mRNA.
- 3 An amino acid was attached to it by the enzyme amino-acyl tRNA transferase.
- 4 It is held in the amino-acyl tRNA binding site of the ribosome.

- A 3 only
 B 4 only
 C 1 and 2
 D 3 and 4

13 Base-pair substitution mutations can have the following molecular consequence except

- A changes one codon for an amino acid into another codon for that same amino acid.
 B codon for one amino acid is changed into a codon for another amino acid.
 C reading frame downstream of the mutation changes
 D codon for one amino acid is changed into a translation termination codon.

- 14** The petals of the morning glory flower are purple when the flower is in bud but become blue as the flower opens. The colour change is caused by an increase in pH of the vacuoles of the petal cells.

A mutation in a gene coding for a type of hydrogen ion (H^+) pump in the vacuole membranes of the petal cells results in flowers that are purple when open.

The difference in DNA sequence of the gene coding for the ion pump in plants with blue or purple flowers is shown below.

plant with blue flowers	-T-T-A-A-T-C-C-T-G-A-G-A-T-T-T-
plant with purple flowers	-T-T-A-A-T-C-C-T-G-C-T-G-A-G-A-T-T-T-

Which statements explain the purple colour of the flowers of the mutant plant?

- 1 A frameshift mutation does not alter the sequence of amino acids in the ion pump.
- 2 The primary structure and the shape of the ion pump are altered.
- 3 The activity of the ion pump is altered by changes in its shape.
- 4 The mutant ion pump cannot pump H^+ ions into the vacuoles of the petal cells

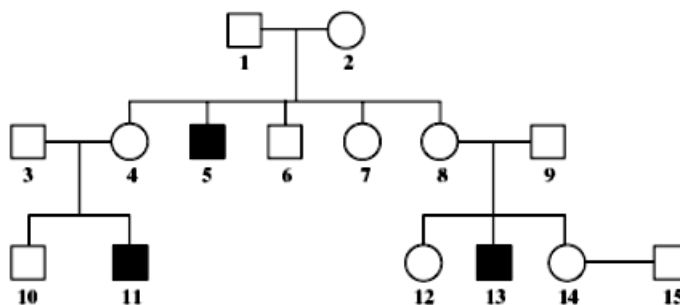
- A** 1 and 2
B 2 and 3
C 3 and 4
D 2, 3 and 4

- 15** A parent organism of unknown genotype is mated in a test cross. Half of the offspring have the same phenotype as the parent.

What can be concluded from this result?

- A** The parent is heterozygous for the trait.
B The parent is homozygous dominant for the trait.
C The parent is homozygous recessive for the trait.
D The trait being inherited is polygenic.

- 16** A genetic disorder causes degeneration of muscle tissue in humans. The diagram shows the inheritance of such a genetic disorder in one family.



Which of the following statements can be concluded based on the diagram above?

- A** Individual 5 was affected with the genetic disorder as a result of genetic mutation during his lifetime.
 - B** Individual 10 should be an affected male too, since his brother (individual 11) was affected with the genetic disorder.
 - C** This genetic disorder is a sex-linked recessive trait.
 - D** This genetic disorder is due to an autosomal dominant allele.
- 17** Which one of the following describes conditions in a photosynthesizing cell, exposed to high light intensity and low carbon dioxide concentration?

	concentration of CO ₂ acceptor	concentration of ATP	concentration of GP
A	high	high	high
B	high	high	low
C	low	high	high
D	low	low	high

- 18** Which of the following statements regarding cyclic and non-cyclic photophosphorylation is true?
- A** Cyclic photophosphorylation only requires light at 680nm to function optimally while non-cyclic photophosphorylation requires light at both 680 nm and 700 nm to function optimally.
 - B** Cyclic photophosphorylation is dependent on products generated through non-cyclic photophosphorylation.
 - C** Only non-cyclic photophosphorylation utilizes electrochemical gradient generated to drive ATP production.
 - D** Only cyclic photophosphorylation can function in the absence of photosystem II.
- 19** Which one of the following substances, when added, would directly result in a rapid decline in ATP production in glycolysis?
- 1 A chemical that would bind to NAD^+ irreversibly and induces its reduction to NADH.
 - 2 An inhibitor that has a similar structure to glucose but cannot be broken down by respiratory enzymes.
 - 3 A chemical that creates an anaerobic environment by reducing the oxygen in the environment
 - 4 A reagent that binds to the active site of ATP synthase permanently
- A** 1 and 2
 - B** 3 and 4
 - C** 1, 2 and 4
 - D** all of the above

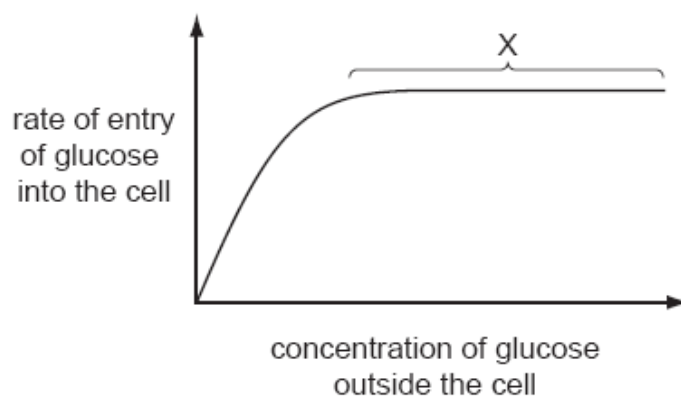
20 The structure of a cell membrane includes these components:

- 1 cholesterol
- 2 glycoproteins
- 3 phospholipids
- 4 proteins

Which component is correctly matched to its function?

	cell-cell recognition	restricting movement of ions across membrane	maintaining stability and fluidity of membrane	transporting polar molecules across membrane
A	1	2	3	4
B	2	3	1	4
C	3	1	4	2
D	3	4	1	2

21 The graph shows how the rate of entry of glucose into a cell changes as the concentration of glucose outside the cell changes.



What is the cause of the plateau at X?

- A All the carrier proteins are saturated with glucose.
- B The carrier proteins are denatured and no longer able to function.
- C The cell has used up its supply of ATP.
- D The concentrations of glucose inside and outside the cell are equal.

- 22 Researchers studying the genetic variation of marine mussels *Mytilus edulis* around Long Island, New York measured the frequency of a particular allele lap^{94} which codes for an enzyme that helps the mussel to regulate its internal saltwater balance.

The data for the above study was presented in the form of the allele frequencies (in percentage) of lap^{94} at 11 different locations. The researchers also noticed that mussel larvae typically swim a long distance away from where they were hatched before they attach themselves to rocks and grow to become adults.

Site	1	2	3	4	5	6	7	8	9	10	11
lap^{94} frequency /%	13	16	16	25	36	37	39	55	59	59	59

A map of Long Island showing the location of the sites and the salinity of the water relative to the distance from the open ocean is shown below.



Using the information from the above table, the map and your biological knowledge, which of the statements below regarding the lap^{94} allele is true?

- 1 Selection against the lap^{94} allele is very strong in the southwest portion of Long Island Sound due to low salinity.
- 2 Selection in favour of the lap^{94} allele is very strong in the northeast portion of Long Island Sound due to the high salinity.
- 3 The frequency of the lap^{94} allele remains unchanged.

- A 1 only
 B 2 only
 C 1 and 2
 D all of the above

- 23** Until the early 1980s, the only species of ground finch living on the small Galapagos island of Daphne Major was *Geospiza fortis*.

In the years after a drought in 1977, when the seeds that they normally ate became scarce, the average beak size of the population became larger, allowing the birds to eat larger, tougher seeds.

In the early 1980s, a larger species of ground finch, *G. magnirostris*, colonised the island and increased in number. *G. magnirostris* has a larger beak than that of *G. fortis*.

The number of both species fell in a drought in 2004, but *G. magnirostris* was better at cracking and eating the remaining large seeds. By 2005, the average beak size of the *G. fortis* population was smaller than in previous years.

Which statements explain the changes in the average beak size of the *G. fortis* population?

- 1 The average beak size in the population changed as a result of natural selection.
- 2 In 1977, *G. fortis* with larger beaks were more likely to survive and breed.
- 3 In 2004, the fitness of *G. fortis* with larger beaks was lowered by the presence of *G. magnirostris*.
- 4 In 2004, *G. fortis* with smaller beaks were more likely to survive and breed.

- A** 1, 2, 3 and 4
B 1, 2 and 4 only
C 2 and 3 only
D 3 and 4 only

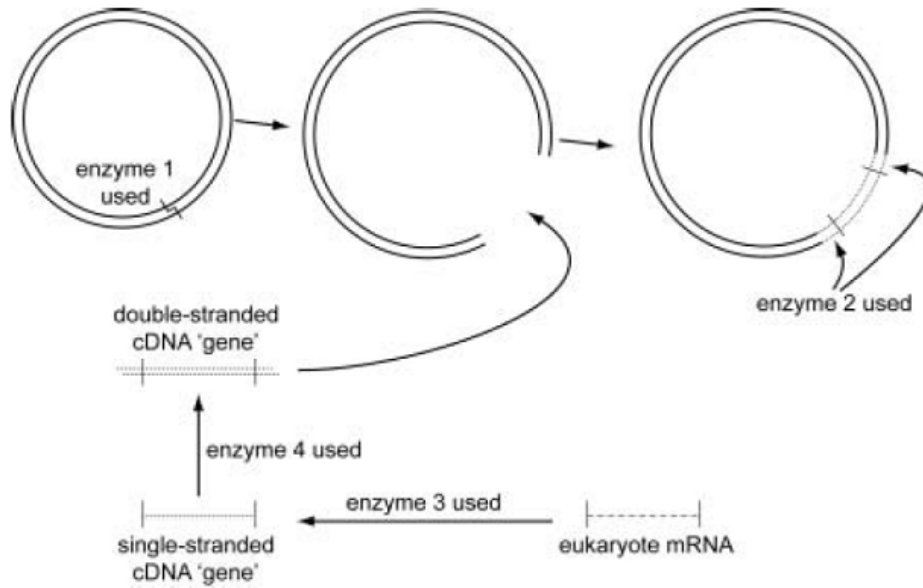
- 24** The table shows the functions of three bones, M, I and S, in three classes of vertebrates. Each bone arises in the same way in embryos of the three classes.

class	M	I	S
fish	part of the articulation of upper and lower jaws	part of the articulation of upper and lower jaws	attaches the upper and lower jaws to the skull
amphibian	part of the articulation of the lower jaw	part of the articulation of the lower jaw	transmits sound to the inner ear
mammals	transmits sound to the inner ear	transmits sound to the inner ear	transmits sound to the inner ear

How do the functions of these bones support Darwin's theory of natural selection?

- A** They show convergent evolution of homologous structures.
- B** They show divergent evolution of analogous structures.
- C** They show how analogous structures have become adapted to perform different functions.
- D** They show how homologous structures have become adapted to perform different functions.

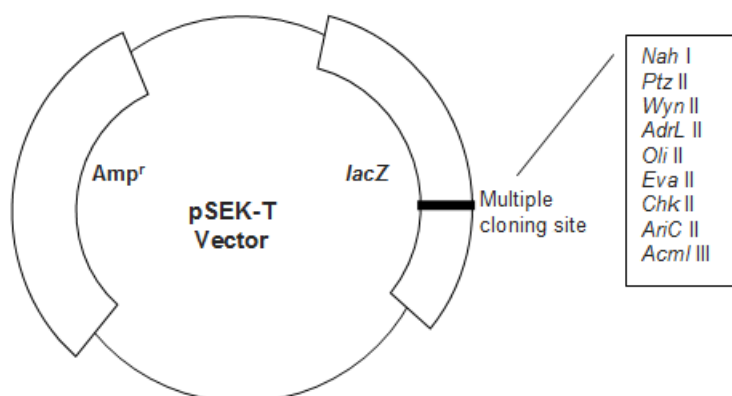
25 The diagram shows part of a process used for genetic engineering.



Which of the following correctly gives the identity of the enzymes used?

	enzyme 1	enzyme 2	enzyme 3	enzyme 4
A	restriction enzyme	DNA ligase	reverse transcriptase	Taq polymerase
B	DNA ligase	restriction enzyme	DNA polymerase	reverse transcriptase
C	restriction enzyme	DNA ligase	reverse transcriptase	DNA polymerase
D	DNA ligase	restriction enzyme	Taq polymerase	reverse transcriptase

- 26 As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the Bt toxin protein into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for Bt toxin were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	chemicals		colour of colonies
A	ampicillin	X-gal	blue
B	ampicillin	X-gal	white
C	β -galactosidase	X-gal	blue
D	β -galactosidase	lactose	white

- 27 The dashed lines in the template sequence represent a long sequence of bases to be amplified.

Template

5' ATTCGGACTTG ----- GTCCAGCTAGAGG 3'

3' TAAGCCTGAAC ----- CAGGTCGATCTCC 5'

Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

- A 3' TAAGCCT 5' & 5' CTAGAGG 3'
- B 5' ATTCGGA 3' & 3' GATCTCC 5'
- C 3' UAAGCCU 5' & 5' CUAGAGG 3'
- D 5' AUUCGGA 3' & 3' GAUCUCC 5'

28 Which of the following are benefits of the Human Genome Project?

- 1 Earlier detection of genetic dispositions to diseases.
- 2 Develop drugs based on individual's genetic makeup.
- 3 Determine genetic variations among different populations of human.
- 4 Establish paternity and other family relationships.

- A** 1 only
B 1 and 2 only
C 2 and 3 only
D All of the above

29 Which of the following statements about stem cells is correct?

- A** Stem cells are specialised cells that have yet to express the genes and proteins characteristic of their differentiated state, and do so when needed for repair of tissues and organs.
- B** Stem cells are fully differentiated cells that reside under the surface of epithelia, in position to take over the function of the tissue when the overlying cells become damaged or worn out.
- C** Embryonic stem cells are undifferentiated cells that are found in all adults, and can give rise to all the cell types in the body.
- D** Stem cells are undifferentiated cells that divide asymmetrically, giving rise to one daughter that remains a stem cell, and one daughter that will differentiate to replace damaged and worn out cells in the adult tissue or organ.

30 Maize varieties are being developed in which the leaves produce proteins that are toxic to insects. The DNA coding for these toxic proteins was inserted into a maize chromosome via a bacterial plasmid. Many people are opposed to this process.

Which objection is **not** biologically valid?

- A** Beneficial insects may be killed if they eat genetically modified maize.
- B** Genes for antibiotic resistance are present in plasmids and these genes may be passed to harmful bacteria.
- C** Hybridisation may transfer the bacterial genes from maize to weeds, giving the weed species new selective advantage.
- D** Mutations may arise in cattle or humans that eat the genetically modified maize.